Nikolay Grantcharov

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EMPLOYMENT

University of Georgia
- Limited-Term Assistant Professor — Aug 2025 – Present

EDUCATION

University of Chicago
- Ph.D. in Mathematics, advised by Victor Ginzburg — Sep 2019 – Aug 2025

University of California, Berkeley
- B.A. in Mathematics — Aug 2015 – May 2019
  - Graduated with highest honors

AWARDS

- Lawrence and Josephine Graves Teaching Prize, University of Chicago — 2025
- NSF Graduate Research Fellowship $\$102,000$ — 2020 – 2023
- Dorothea Klumpke Roberts Prize in Mathematics, UC Berkeley — 2019
  UC Berkeley department prize awarded to seniors who have demonstrated truly exceptional scholarship in mathematics
- UC Berkeley Regents' and Chancellor's Scholarship, $\$10,000$ — 2015 – 2019
  Awarded to 200 undergraduates each year, most prestigious scholarship offered by UC Berkeley.
- UC Berkeley Dean's Honors List — 2015 2017
  Top 4% of UC Berkeley College of Letters and Science Undergraduates.

PUBLICATIONS

Cohomology and Representation Theory for Lie Superalgebras — October 2026
- With D. K. Nakano. arXiv:2610.25642. Submitted.

Cohomology on Cotangent Bundles of Partial Flag Varieties in Type A — September 2026
- With an appendix joint with T. Gannon. arXiv:2609.04670. Submitted.

Deformation Theory of the $(C_n^\vee, C_n)$-DAHA — June 2026
- arXiv:2606.29327. Submitted.

Quasi-Classical Braverman–Kazhdan Intertwiners via Quiver Varieties — June 2026
- With A. Slipper. arXiv:2606.14637. Submitted.

Infinitesimal jet spaces of Bun_G in positive characteristic — 2024
- arXiv:2402.10369. Submitted.

On BBW Parabolics for Simple Classical Lie Superalgebras — 2021
- With D. Grantcharov, D. K. Nakano, and J. Wu. *Advances in Mathematics*, **381** (2021), 107647, 44 pages. arXiv:1810.06980.

Extension Quiver for Lie Superalgebra $\mathfrak{q}(3)$ — 2020
- With V. Serganova. *SIGMA*, **16** (2020), 141, 32 pages.

TEACHING

University of Georgia Instructor — Aug 2025 – Present
- Spring 2027 (scheduled): Two sections of Calculus III (Math 2270).
- Fall 2026: One section of Calculus I (Math 2250).
- Spring 2026: One section of Calculus I (Math 2250).
- Fall 2025: Two sections of Calculus II (Math 2260).

University of Chicago Instructor and Teaching Assistant — 2021 – 2025
- Spring 2025: Teaching Assistant for the Math in Paris Study Abroad Program.
- Fall 2024: Math 133: Calculus III.
- Fall 2021 – Spring 2022: Math 130s – Calculus I, II, and III.

University of Chicago College Fellow Sep 2020 – Jun 2021

- Spring 2021: Math 259 Honors Basic Algebra III.
- Winter 2021: Math 244 Introduction to Algebraic Geometry.
- Fall 2020: Math 254 Basic Algebra I.

AwesomeMath Summer Program Instructor Jun 2017 – Present

- AwesomeMath is a three-week summer program for gifted high school students training for math olympiads such as USAMO and IMO.
- Taught intermediate and advanced Olympiad courses in Euclidean geometry and algebra every summer since 2017.
- Helped develop the curriculum and course notes for intermediate and advanced geometry courses.

SEMINARS ORGANIZED

The Bun_G Seminar 2023 – 2025

- Co-founded the Bun_G seminar with Aaron Slipper in January 2023 and organized it for seven consecutive quarters. Across more than 50 talks, the seminar explored the geometry and representation theory of the moduli stack of principal bundles on a curve and its role in the geometric Langlands program, with talks by leading faculty.

UGA Algebra Seminar Aug 2025 – Present

- Co-organizer, University of Georgia Algebra Seminar.

AMS Southeastern Sectional Meeting, Kennesaw State University October 10–11, 2026

- Co-organizer, Special Session on Geometric and Homological Methods in Representation Theory.

WORK EXPERIENCE

UChicago Graduate Algebra Teaching Assistant Sep 2022 – Jun 2024

- TA and HW grader for graduate Algebra I Fall 2022 and Fall 2023.
- TA and HW grader for graduate Algebra III Spring 2024.

UChicago Math DRP Mentor Sep 2022 – Sep 2023

- Mentored C. Chang in a representation theory for finite groups and Lie algebras reading course

UChicago REU Mentor Jun 2020 – Aug 2020

- Helped run the apprentice program and mentored J.Yang on project "Super five of Ramsey Theory" and G. Graham "The ring of symmetric polynomials"

Berkeley Math Circle Assistant Jul 2015 – Jul 2016

- Managed BMC Website (http://mathcircle.berkeley.edu).
- Handled email communication with BMC instructors, parents and students.
- Helped with selection process for 400 applicants.

Berkeley Math Homework Grader Jan 2017 – May 2019

- Grade weekly homework assignments for Abstract Algebra (113) for 40 students in the following terms: Spring 2017, Summer 2017, Fall 2017, Spring 2018, Spring 2019.
- Grade for graduate level Commutative Algebra (250B) Spring 2019 term.

AwesomeMath Summer Program Teaching Assistant Jun 2014 – Aug 2016

- In charge of problem sessions, held office hours, and graded exams for two courses per camp.
- Throughout 7 camps, I was a TA for: Algebra 2.5, 3.5; Geometry 1.5, 2.5; Number Theory 2.5.

SEMINAR AND CONFERENCE TALKS

Forthcoming

- AMS Sectional Meeting, Syracuse, New York, "Braverman–Kazhdan intertwiners via quiver varieties", April 11, 2027.
- Algebra Seminar, Georgia Tech, November 30, 2026.
- Algebra Seminar, University of Pennsylvania, November 6, 2026.

2026

- AMS Sectional Meeting, Kennesaw, Georgia, "Braverman–Kazhdan intertwiners via quiver varieties", October 11, 2026.
- Geometry, Physics, and Representation Theory Seminar, Northeastern University, September 23, 2026.
- Representation Theory Seminar, University of North Carolina, Chapel Hill, August 28, 2026.
- ICM satellite conference, University of Virginia, "Braverman–Kazhdan intertwiners via quiver varieties" (contributed talk), July 20, 2026.
- Algebra Seminar, University of California, Riverside, "Symmetries of the cotangent bundle of $SL_n/[P,P]$", May 18, 2026.

- Informal Mathematical Physics Seminar, Columbia University, "Symmetries of the cotangent bundle of $SL_n/[P,P]$", March 26, 2026.

2025

- Algebra Seminar, UGA, "Symmetries of the multiplicative cotangent bundle of G/U, parts I and II", September 8 and 15, 2025.
- Coulomb Branch Seminar, Northwestern University, March 10, 2025.
- UChicago Bun_G Seminar, "Geometric realizations of the affine Hecke algebra", January 15, 2025.

2024

- Geometry and Physics Seminar, Northwestern University, "Infinitesimal structure of Bun_G", December 5, 2024.
- Algebra Seminar, UGA, "Infinitesimal structure of Bun_G", November 11, 2024.
- Mathematical Physics Seminar, LSU, "Infinitesimal structure of Bun_G", October 28, 2024.
- UChicago Bun_G Seminar, "Representation theory for finite W-algebras", October 16, 2024.
- Mathematical Physics Seminar, Perimeter Institute, "Infinitesimal structure of Bun_G", May 23, 2024.
- UChicago Bun_G Seminar, "The global Hecke action", May 1, 2024.
- UChicago Bun_G Seminar, "Non-abelian Poincaré duality", February 14, 2024.

2023

- UChicago Bun_G Seminar, "The local and global Hitchin map", October 18, 2023.
- AMS Fall Sectional Meeting, Mobile, Alabama, "Infinitesimal neighborhoods of Bun_G in positive characteristic", October 14, 2023.
- University of Bath Representation Theory Seminar, "Cherednik algebras and their representation theory" (four-hour lecture series), August 14–15, 2023.
- AMS Spring Sectional Meeting, Cincinnati, Ohio, "Infinitesimal neighborhoods of Bun_G in positive characteristic", April 15, 2023.
- UChicago Bun_G Seminar, "Infinitesimal neighborhoods of Bun_G", March 29, 2023.

2022

- Canada–Mexico–USA Conference in Representation Theory, Noncommutative Algebra, and Categorification, "On BBW parabolics for Lie superalgebras" (poster), June 9–12, 2022.
- UChicago Student Representation Theory Seminar, "A compactification of a configuration space for curves", May 22, 2022.
- Workshop on Supergeometry, Fields Institute, Toronto, Ontario, "Finite-dimensional representations of the queer Lie supergroup $Q(n)$", March 21, 2022.

2021

- UChicago Student Representation Theory Seminar, "Haiman's proof of the $n!$ conjecture, parts I and II", October 20 and 27, 2021.
- Southeast Lie Theory XII, Charleston, South Carolina, "BBW parabolics for classical Lie superalgebras", October 16, 2021.
- Columbia University Student Seminar on Category $\mathcal{O}$ (online), "Projective functors", April 2, 2021.
- Columbia University Student Seminar on Category $\mathcal{O}$ (online), "Parabolic category $\mathcal{O}$", February 12, 2021.

2020

- Superalgebra Theory and Representations Seminar (online), Weizmann Institute, "Finite-dimensional representation theory of the queer Lie superalgebra $\mathfrak{q}(n)$", December 2, 2020.
- Columbia University Student Seminar on Category $\mathcal{O}$, "Translation functors for BGG category $\mathcal{O}$", September 18, 2020.

2019

- Representation Theory and Integrable Systems conference, Zurich, Switzerland, “Extension quiver for the Lie superalgebra $\mathfrak{q}(3)$”, August 14, 2019.
- Math Monday Undergraduate Seminar, UC Berkeley, “Symmetric polynomials and representation theory”, April 22, 2019.
- Global Langlands over Function Fields Seminar, UC Berkeley, “Excursion operators”, April 4, 2019.

CV updated on 2026-10-25